Xiaokun (Ken) Zhong

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Work Authorization: U.S. Permanent Resident (EAD, I-485 filed). No visa sponsorship required.

Education

University of California, Berkeley Dec 2025 – May 2026
Visiting Student, Computer Science & Mathematics *Berkeley, CA*

Hong Kong University of Science & Technology (HKUST) Sep 2022 – May 2026
BSc in Mathematics (Computer Science Track) *Hong Kong*

- Coursework: Differential Equations, Artificial Intelligence, Probability Theory.

Research & Engineering Experience

Dartmouth College & UC Berkeley Feb 2025 – Present
Research Engineer, AI Systems & Scientific ML *Berkeley, CA*

- Investigated failure modes and training instability in scientific AI systems under challenging regimes (e.g., stiff PDE settings), identifying factors affecting convergence and generalization.
- Designed and implemented second-order optimization pipelines (Newton-CG) in PyTorch, improving solution accuracy by up to 50% and enabling stable training across benchmark tasks.
- Built scalable ML experimentation infrastructure on Slurm-based HPC clusters, enabling large-scale ablation studies and reproducible evaluation of optimization strategies.
- Led research on regime-specific optimization for scientific ML, resulting in a co-first author submission to ICML 2026.

HKUST Undergraduate Research Program Feb 2023 – May 2025
Research Engineer, Medical Imaging AI *Hong Kong*

- Developed deep learning models for MRI reconstruction and 3D volumetric modeling from sparse medical data.
- Improved data processing and training pipelines in Python and C++, enabling more efficient experimentation on large-scale imaging datasets.

Publications

Y. Wang*, Y. Hu*, **X. Zhong*** et al. "Unveiling Multi-regime Patterns in SciML: Distinct Failure Modes and Regime-specific Optimization." **ICML 2026**. arXiv:2605.29153.

Technical Skills

Machine Learning: PyTorch, scientific machine learning (PINNs, Neural ODEs), optimization, training dynamics

Systems & Infrastructure: Linux, Slurm, Docker, HPC workflows, distributed experiments

Languages: Python, C++, SQL, MATLAB

Selected Projects

Secure Network Tunneling & Transport Optimization Nov 2025
Systems Project *Remote*

- Designed and deployed a secure cross-region networking system on VPS infrastructure to improve reliability and latency under constrained conditions.
- Implemented TLS-based transport and system-level optimizations, analyzing trade-offs between throughput, latency, and network stability.

Autonomous Robotics Vision System (RoboMaster) Oct 2022 – Jun 2023
Computer Vision Engineer *Hong Kong*

- Built real-time computer vision pipelines for autonomous target acquisition using OpenCV and Python.
- Implemented C++ drivers for sensor fusion and motor control on embedded STM32 platforms.

Industry Experience

Agricultural Bank of China Jan 2024 – Feb 2024
IT Intern (e-CNY Digital Currency) *Shenzhen, China*

- Tested and improved security and functionality of an internal e-CNY wallet system, contributing to a large-scale digital currency pilot.